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Experiment 07

VLAN Configuration, Router-on-a-Stick & Multi-VLAN DHCP Server

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Aim

To implement a router as a DHCP server that can serve multiple VLANs, using Cisco Packet Tracer.

Learning Objectives

After completing this experiment, we should be able to:

1. Understand the concept of VLANs.
2. Create and configure multiple VLANs on a switch.
3. Assign switch ports to appropriate VLANs.
4. Understand the difference between an Access Port and a Trunk Port.
5. Configure a trunk between a switch and a router.
6. Understand Inter-VLAN Routing and the Router-on-a-Stick technique.
7. Configure router subinterfaces for multiple VLANs.
8. Configure a router as a DHCP Server with separate DHCP pools per VLAN.
9. Verify VLAN, trunk, DHCP, and routing configurations.

Theory

VLANs and Broadcast Domains

A **VLAN** (Virtual Local Area Network) is a logical network created within a physical switch. Instead of placing every connected device into the same broadcast domain, a switch's ports can be logically grouped into separate VLANs (for example, one VLAN for students and another for faculty), even though every device shares the same physical switch. This gives better network segmentation, improved security, a smaller broadcast domain per group, and easier management.

Inter-VLAN Routing and Router-on-a-Stick

Each VLAN is a separate Layer 2 network, so a switch alone cannot forward traffic between two different VLANs – a Layer 3 device (a router or a Layer 3 switch) is required. This process is called **Inter-VLAN Routing**. **Router-on-a-Stick** performs inter-VLAN routing using a single physical router interface that is divided into multiple logical **subinterfaces** – one per VLAN – so that one physical port can serve every VLAN instead of needing a dedicated cable per VLAN. Each subinterface is assigned an IP address that acts as the default gateway for its VLAN.

VLAN Tagging (IEEE 802.1Q)

When several VLANs share a single trunk link, the receiving device needs to know which VLAN each frame belongs to. This is achieved with **802.1Q VLAN tagging**, which inserts a VLAN identifier into the Ethernet frame as it crosses the trunk. On the router, the command



`encapsulation dot1Q <vlan-id>` tells a subinterface which VLAN's tagged frames it should handle.

DHCP and the DORA Process

DHCP (Dynamic Host Configuration Protocol) automatically assigns network configuration – IP address, subnet mask, default gateway, and DNS server – to client devices, removing the need to configure every host manually. A Cisco router can itself act as a DHCP server. The DHCP exchange follows four steps, known as **DORA**:

- **D – Discover:** the client broadcasts a request asking if a DHCP server is available.
- **O – Offer:** the server offers an available IP address.
- **R – Request:** the client requests the offered address.
- **A – Acknowledgment:** the server confirms the allocation.

When one router serves multiple VLANs, it maintains a **separate DHCP pool per VLAN**, each pool tied to that VLAN's subnet, so that VLAN 2 clients receive addresses from the VLAN 2 pool and VLAN 3 clients receive addresses from the VLAN 3 pool. The address already used by the router's own subinterface (the gateway) is marked with `ip dhcp excluded-address` so that DHCP never hands it out to a client.

Network Topology

The router (Router0) connects to the switch (Switch0) over a single trunk link carrying both VLANs; the switch's remaining ports are split between the two VLANs, two access ports each. The *last* port of the switch (Fa0/24) is deliberately used for the router link, keeping it visually and logically separate from the numbered access ports (Fa0/1--Fa0/4) used by the PCs – this makes it easy to tell the access ports and the trunk port apart at a glance.

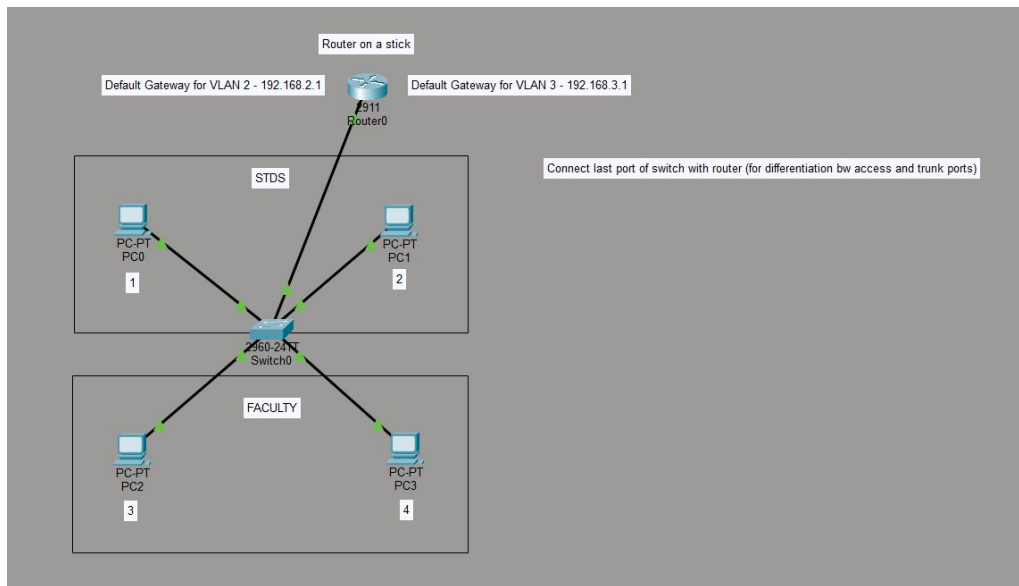
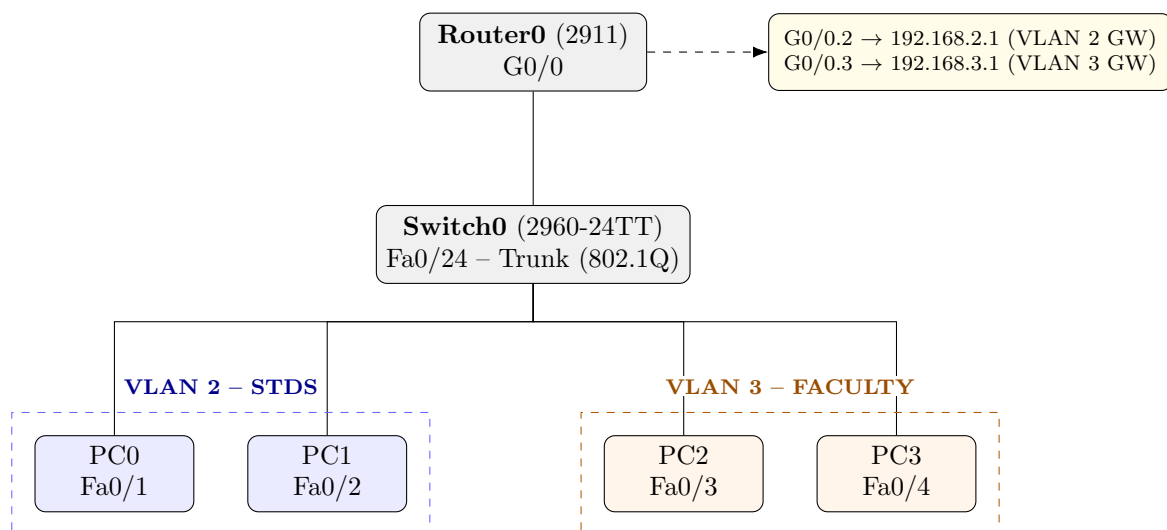


Figure: Topology as built in Cisco Packet Tracer.

The same topology is redrawn below as a labelled logical diagram, showing the VLAN grouping and the router's subinterface addressing more clearly.



Devices Used

1 Router (2911) – Router0; 1 Switch (2960-24TT) – Switch0; 4 PCs – PC0, PC1, PC2, PC3.
All PC-to-switch and switch-to-router links use Copper Straight-Through cables.



VLAN and Addressing Plan

VLAN	Name	Network	Gateway
2	STDS	192.168.2.0/24	192.168.2.1
3	FACULTY	192.168.3.0/24	192.168.3.1

Device / Port / VLAN Mapping

Device	Switch Port	VLAN
PC0	Fa0/1	VLAN 2
PC1	Fa0/2	VLAN 2
PC2	Fa0/3	VLAN 3
PC3	Fa0/4	VLAN 3
Router0	Fa0/24	Trunk (802.1Q)



Procedure

Part A – Configure the Switch (Switch0)

Step 1 – Enter privileged and global configuration mode

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

Step 2 – Create VLAN 2 and VLAN 3

```
Switch(config)# vlan 2
Switch(config-vlan)# name STDS
Switch(config-vlan)# exit

Switch(config)# vlan 3
Switch(config-vlan)# name FACULTY
Switch(config-vlan)# exit
```

Step 3 – Verify that the VLANs were created

```
Switch(config)# do show vlan brief
```

Step 4 – Assign PC0 and PC1 (Fa0/1–Fa0/2) to VLAN 2

```
Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 2
Switch(config-if-range)# exit
```

Step 5 – Assign PC2 and PC3 (Fa0/3–Fa0/4) to VLAN 3

```
Switch(config)# interface range fastEthernet 0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 3
Switch(config-if-range)# exit
```

Step 6 – Configure Fa0/24 as a trunk port (link to the router)

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

Step 7 – Verify the trunk

```
Switch# show interfaces trunk
```

Fa0/24 should now be listed as a trunk port, confirming the switch is ready to carry both VLANs toward the router.



Part B – Configure the Router (Router0)

Step 1 – Enter privileged and global configuration mode

```
Router> enable
Router# configure terminal
Router(config)#
```

Step 2 – Bring up the physical interface G0/0

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 3 – Create the subinterface for VLAN 2

```
Router(config)# interface gigabitEthernet 0/0.2
Router(config-subif)# encapsulation dot1Q 2
Router(config-subif)# ip address 192.168.2.1 255.255.255.0
Router(config-subif)# exit
```

Step 4 – Create the subinterface for VLAN 3

```
Router(config)# interface gigabitEthernet 0/0.3
Router(config-subif)# encapsulation dot1Q 3
Router(config-subif)# ip address 192.168.3.1 255.255.255.0
Router(config-subif)# exit
```

Step 5 – Verify the router interfaces

```
Router# show ip interface brief
```

Both G0/0.2 and G0/0.3 should show as up/up, with 192.168.2.1 and 192.168.3.1 respectively. This is **Router-on-a-Stick**: one physical interface, two logical subinterfaces, each acting as the gateway for its own VLAN, connected to the switch over a single trunk link.

Step 6 – Exclude the gateway addresses from DHCP

```
Router(config)# ip dhcp excluded-address 192.168.2.1
Router(config)# ip dhcp excluded-address 192.168.3.1
```

Step 7 – Create the DHCP pool for VLAN 2

```
Router(config)# ip dhcp pool VLAN2_POOL
Router(dhcp-config)# network 192.168.2.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.2.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

Step 8 – Create the DHCP pool for VLAN 3

```
Router(config)# ip dhcp pool VLAN3_POOL
```



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```
Router(dhcp-config)# network 192.168.3.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.3.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

Part C – Configure the PCs

On each PC, go to **Desktop** → **IP Configuration** and select **DHCP** instead of entering an address manually. PC0 and PC1 (VLAN 2) should automatically receive addresses in 192.168.2.0/24; PC2 and PC3 (VLAN 3) should receive addresses in 192.168.3.0/24.



Verification / Output

show vlan brief (on Switch0)

VLAN	Name	Status	Ports
1	default	active	
2	STDS	active	Fa0/1, Fa0/2
3	FACULTY	active	Fa0/3, Fa0/4

show interfaces trunk (on Switch0)

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/24	1,2,3			

show ip interface brief (on Router0)

Interface	IP-Address	Status	Protocol
GigabitEthernet0/0	unassigned	up	up
GigabitEthernet0/0.2	192.168.2.1	up	up
GigabitEthernet0/0.3	192.168.3.1	up	up

IP Configuration received by each PC (via DHCP)

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.2.2	255.255.255.0	192.168.2.1
PC1	192.168.2.3	255.255.255.0	192.168.2.1
PC2	192.168.3.2	255.255.255.0	192.168.3.1
PC3	192.168.3.3	255.255.255.0	192.168.3.1

Note: the exact addresses assigned can vary slightly depending on lease order in Packet Tracer; the important part is that each PC lands in the correct VLAN's subnet with the correct gateway.

show ip dhcp binding (on Router0)

IP address	Client-ID/Hardware address	Lease expiration
192.168.2.2	0100.5079.6668.0a	...
192.168.2.3	0100.5079.6668.0b	...
192.168.3.2	0100.5079.6668.0c	...
192.168.3.3	0100.5079.6668.0d	...

Result

The switch was configured with two VLANs (STDS and FACULTY) and a trunk link to the router. Using Router-on-a-Stick, the router's single physical interface was split into two subinterfaces, one per VLAN, each acting as that VLAN's default gateway. The router was then configured as a DHCP server with a separate address pool per VLAN, so that PC0/PC1 automatically obtained addresses in 192.168.2.0/24 and PC2/PC3 automatically obtained addresses in 192.168.3.0/24 – confirming that a single router can simultaneously provide inter-VLAN routing and DHCP services to multiple VLANs.

Learning Outcome

After completing this experiment, the following learning outcomes were achieved.

- Understood how VLANs logically segment a single physical switch into separate broadcast domains.
- Learned to create VLANs and assign access ports to them using `switchport access vlan`.
- Understood the difference between an access port (carries one VLAN, untagged) and a trunk port (carries multiple VLANs, 802.1Q-tagged).
- Implemented Router-on-a-Stick using subinterfaces and `encapsulation dot1q` to perform inter-VLAN routing over a single physical link.
- Configured a Cisco router as a DHCP server with per-VLAN address pools, excluded gateway addresses, and default-router/DNS options.
- Verified the configuration end-to-end using `show vlan brief`, `show interfaces trunk`, `show ip interface brief`, and `show ip dhcp binding`.
- Gained practical, hands-on exposure to building and testing a multi-VLAN, DHCP-enabled network in Cisco Packet Tracer.